



Multi-objective approach for protection of microgrids using surrogate assisted particle swarm optimization (SAPSO)

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Abstract

Despite of several formulations discussed for operation of Directional Over current Relays (DOCRs) in micro grids the coordination of DOCR is yet a great task to be solved. A micro grid is a group of generators preferably a renewable energy source and loads interconnected. The modes of micro grid operation, grid connected mode and autonomous mode are to be considered simultaneously to design an optimal protection scheme for the micro grid. Several conflicting objective functions that must be optimized simultaneously during the relay setting process make the coordination problem to be treated as multi-objective. In this work a multi-objective approach yielding better coordination is presented. There are two major objectives as follows; former is the synchronization period between the main and secondary relays, later is the relay operation time. The conventional Multi-Objective Evolutionary Algorithms (MOEAs) may take thousands of function evaluations, which is not practical for real-world problems. Hence surrogates are introduced in those MOEAs to approximate the fitness function. The relay coordination is devised as a constrained multi-objective problem. The Surrogate Assisted Particle Swarm Optimization (SAPSO) Algorithm is applied to accomplish better coordination between the relays. The scheme is validated using standard IEEE systems and the results are compared with the conventional Multi-Objective Particle Swarm Optimization (MOPSO). The optimal values of pickup current and Time Dial Settings are computed for both modes of micro grid operation. The sum of operating time of all relays and the synchronization period between the main and secondary relays are minimized. The Inverted Generation Distance (IGD) is employed as the parameter for computing the performance of the algorithm. Also the convergence characteristic is analyzed. From the results it is proved that SAPSO have good stability than MOPSO and offers better Pareto fronts, convergence and distribution.

Keywords Micro grids · Relay coordination · Operating time · CTI and surrogates

Introduction

The need for electric power is growing rapid in the current scenario. The conventional energy sources known as non renewable source takes millions of years for their formation.

The availability of these resources is depreciated. To overcome these, new techniques for the production of electricity is proposed to meet the demand of the electricity with the advancement in the world. The concept of Micro grid is a solution to generate and distribute power which eliminates

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the conventional resources for power generation. A micro grid is a group of generators preferably a renewable energy source and loads interconnected, that can be operated in connection with the grid or operated autonomously ensuring reliable supply to the customer. For reliable operation, i.e., to minimize transmission losses and network congestion, these micro grids should be connected near the load. The micro grid consists of Distributed Energy Resources (DERs) and energy storage devices. The advantages of micro grids are energy efficient, reliable, economic, eco-friendly and reduced energy consumption. Under normal condition the micro grids are connected to the distribution network. Under abnormalities it can be separated from the grid by static switch at the Point of Common Coupling (PCC) and operate autonomously.

Even so, there are many technical issues during the integration and operation of micro grids (Najy Waleed et al. 2013; Haron et al. 2012; Sharaf et al. 2016; Bayindir et al. 2014; Olivares et al. 2014). The voltage and frequency of the system is affected by the intermittent nature of DG. The variation in the load results in un-modeled dynamics. This un-modeled dynamics has impacts on the performance of the micro grid. The security, autonomous operation, recovery from planned and unplanned shut downs, relay coordination, reliability and disturbance in quality of power, cost incurred for the modern technologies are the vital issues need to be addressed in the micro grids.

One among the challenges in the integration of micro grid is the protection system. Distribution systems need to be protected from all the possible faults. When a fault occurs only the portion of the system affected by the fault should be tripped, while the other healthy areas should be left unaffected. Protection coordination is designing protection scheme such that each relay operates as rapid as possible, but if the relay fails, the backup relay comes into action. In Conventional distribution systems, the power flow is unidirectional hence the protection system is simple and so over current relays and fuses are installed to protect the system (Elmore 2003; Singh and Panigrahi 2014; Zeineldin et al. 2006; Asadi and Kouhsari 2009). Due to the new concept of introducing generation in the distribution side (DER), the power flow is no longer unidirectional but becomes bidirectional. This makes the protection scheme and hence the coordination of relays becomes complex.

Also introduction of DER affects the magnitude and direction of fault current. Hence the relays cannot perform their function correctly and there is a chance of miss-coordination between the relays. The fault current magnitude depends on the type of DG. For inverter-based DG, the fault current value is approximately 1 p.u. The synchronous based DG has more values. The introduction of DER into the system makes the system more interconnected. Therefore the DOCR are implemented to solve the coordination problem (Chen et al. 2013; El-Khattam and Sidhu 2008; Sortomme et al. 2009; Javadian

and Haghifam 2008; Kamal et al. 2014). The abnormalities such as line outage, generator outage and substation faults, etc., may lead to miss-coordination of DOCR because variation in fault current cases. During failure of the supply, the micro grid is changed from grid connected to the autonomous mode and the network topology changes. The DOCR sense the current and voltage of the system and trip the relay if the fault is detected. DOCR determines the fault by considering the phase or of voltage and current. The relay finds the location of fault. If the fault is located in front of the relay, then it will operate, otherwise it does not operate. The relay operates by comparing current in the circuit with the reference current. When the actual current exceeds the reference current; it is called short circuit current. This is used to identify the location of fault. The magnitude of the fault current is inversely proportional to the fault location. If the fault is located far away from the source, the fault impedance will be high and therefore the fault current is small.

In protection system, the relays are considered as primary and backup relays. The relay responding fast to the faults is termed as primary or main relay. The relay operating after a short time interval when the primary relay fails is considered as backup relay. Since the main relay has number of backup relays for protection, the constraints imposed are relatively high. In event of this, the relay coordination in meshed power system is a great challenge for the researchers (Mohammadi et al. 2010).

The DOCR coordination issue is typically solved as a single objective optimization problem considering only one objective function. In general, the over current relay setting is proposed as several contrariant functions to be refined simultaneously. The coordination of DOCR coordination should conform constraints imposed by manufacturers such as the electrical and mechanical damage limits. Also the operation characteristics of the equipments such as the transformer inrush current, motor locked rotor current should be fulfilled. The incorporation of renewable sources as generators makes the DOCR coordination problem intricate compared with radial power system due to the following reasons: the current direction and the short circuit components. The relay operating time is minimized by choosing optimal relay design variables such as pickup (PU) and time Multiplier Setting (TMS) settings of relay. The problem is formulated as multi-objective optimization problem as a new objective function which is the minimization of CTI, is added to the primary relay operating time function.

The advantage of the multi-objective optimization is simultaneous optimization of the two conflicting objectives (Benabid 2016; Díaz-Manríquez et al. 2016; Lalitha and Xavier 2019; Hussain et al. 2001; Liu and Yang 2012). Conventional optimization techniques such as Newton method, steepest descent are used to solve many engineering optimization problems. But they have the limitations, that they are

not able to find a global solution in case of problems with multiple local minima. Hence the conventional optimization algorithms are replaced by global optimization algorithms.

The global optimization algorithms may be heuristic and meta-heuristic in nature. These algorithms are easy to implement and also efficient in nature. The types of nature inspired algorithms are GA, PSO, and Bacterial foraging algorithm, Cuckoo search algorithm, Firefly algorithm, Cat swarm optimization etc. These algorithms are not knowledge based and provide better accurate results (Yang and Liu 2013; Mansour et al. 2007; Nigim and Lee 2007; Faisal Badal et al. 2019; Singh and Gupta 2012; Barzegari et al. 2010; Zellagui et al. 2014; Yu et al. 2018). Even though these optimization algorithms are useful to solve complex optimization algorithms, they are incapable when the objective function is computationally complex and expensive, since it requires high number of iterations and function evaluations. The time taken to complete first iteration is somewhat high in these types of nature inspired evolutionary algorithms and it is not applied to real-world optimization problems with large number of function evaluations. Hence nowadays, surrogate assisted optimization algorithms are used to solve the computationally expensive real-world optimization problems. These surrogate model are classified as Radial basis function (RBF), Neural networks, Kriging, etc. These reduce the complexity by approximating the function and thereby reducing the number of function evaluations and computational cost. These surrogate models will assist the evolutionary optimization algorithms to converge quickly. In this work, Surrogate Assisted Particle Swarm Optimization (SAPSO) is proposed for solving the multi-objective optimization problem in DOCR relay coordination (Korde and Bedekar, 2016; Amali and Baskar 2015a; Buche et al. 2005). The problem considered is multi-objective, since the objective function needs to be minimized comprises of both relay operating time and coordinating time (Lv et al. 2019; Ratle 2001; Cai et al. 2019; Korde and Bedekar. 2016; Amali and Baskar 2015; Buche et al. 2005). The results are obtained for both grid connected configuration and double configuration. The simulation is performed by inducing faults in the middle of lines. Each relay operates for close end and far ends faults. In Grid connected micro grid configuration the system is connected to grid and the analysis is performed. The system is supplied from the grid and also from all the DG. The system is said to be islanded when there is no supply from the grid and is powered only by DGs.

Related work

The integration of micro grids its challenges and the control methods for micro grid operation have been proposed in Faisal et al. (2019). The voltage and frequency of the system is affected by the intermittent nature of DG. The voltage

of the micro grid is regulated with the implementation of integral controller, PID controller and damping controllers. Also the variations in the load may result in un-conditioned dynamics. This dynamics effect the microgrid operation. The Micro grid is controlled by a central controller to estimate the amount of electric power supplied by the DG units. A decentralized control approach in which each unit has a local controller is also discussed. With these controllers desirable performance of micro grid is obtained. The opportunities and challenges faced while the integration of micro grid with existing utilities are discussed in Nigim et al. (2007). The coordination of DOCR coordination should conform constraints imposed by manufacturers such as the electrical and mechanical damage limits. Also the operation characteristics of the equipments such as the transformer in rush current, motor-locked rotor current are discussed. The incorporation of renewable sources as generators makes the DOCR coordination problem intricate compared with radial power system due to the following reasons: the current direction and the short circuit components The electric power pricing, incentives to effective utilization of power, decision priority, risk responsibility and insurance for new technologies adaptation, interconnection standards and regulatory control are analyzed. The reliance of import of fuel sources, efficient use of RES, electric power supplies to rural region, extension of transmission lines and career growth in modern technology areas are also presented.

Numerous literatures have been proposed for relay coordination problem. Over current protection issues in a micro grid have been presented Waleed et al. (2013). The solutions include use of Fault Current Limiters (FCL), directional protection and designing the relays with two settings, one for grid operated mode and another for autonomous mode. The relay settings are computed considering the both modes of micro grids. The solution for the problem is obtained using constrained non linear programming and the optimization is carried out using Genetic Algorithm. The protection scheme can be implemented to meshed distribution system with conventional synchronous generator. The FCL is connected in series with the main grid and along with the DOCR, to offer an efficient protection system. Over current protection relay operates when the load current exceeds the limit. The Tap Setting (TS) is the term referred as the current setting multipliers in over current relays which is normally 50–200% increase Korde and Bedekar (2016). The maximum load current and minimum fault current can be used for measuring the TS Mohammadi et al. (2010). If a fault occurs in the system the relay should operate fast to clear the fault. If the relay fails, the backup relay should come into action after a time interval called Coordination Time Interval (CTI) Eklas Hossain et al. (2014). The relay settings are mainly influenced by initial guess of the relay parameters. The relay coordination may depend on time grading, current

grading or both time and current grading (So et al. 1997). The optimum PSM and TMS and minimum operation time of primary relay are computed. Fault Current Limiters are implemented to reduce the DG fault current in El Khattam and Sidhu (2008). The relay coordination is restored without varying the relay settings. Also the analysis is carried out without disconnecting the DG at the time of fault.

The steps to select the resistive or inductive FCL are proposed. The system is analyzed without DG, implementation of DG with FCL and in the absence of FCL. In Manohar Singh and Panigrahi (2010) the operating time gap of all key relays in the faulted line is minimized. The selectivity constraints are added to the existing over current relay constraints. To accelerate the convergence, Evolution-based Differential Search Algorithm is used to minimize the operating time. Intensive researches have been carried out for relay coordination issues using several evolutionary-based optimization tools. Particle Swarm Algorithm (PSO) Zeineldin et al. (2006), Genetic Algorithm (GA) Singh et al. (S 2012), Harmony Search Algorithm (HSA) and Artificial Bee Colony (Uthitsunthorn and Kulworawanichpong 2012) for relay coordination have been presented. In Singh et al. (2012) the relay selectivity without affecting the sensitivity and fault clearing time is achieved for solving coordination of Directional over current relay. Zellagui et al. (2014) applied Firefly Algorithm (FA) to find the TMS and pickup current (I_p) so that the operation time of relay is reduced. Here the objective function is single. The inverse definite Minimum time directional over current relay is implemented. The relay parameters are computed in consideration with the change in impedance of the line due to series compensation. The relay settings with and without series compensation are compared and discussed. In Hussain et al. (2001) multi-objective approach is used to resolve the coordination of DOCRs using Modified Firefly Algorithm (MFA). The optimization is performed to yield Time Dial Settings. The pickup currents are computed depending on the load current. The research reveals that the population size has the impact on the computation time.

In Benabid (2016), the time of operation of all primary relay and the operating time of back up relay are the two objectives to be minimized simultaneously. The Time Dial Settings are taken as real variables and the pickup current are taken as integer variables. To balance the real and integer variables, the enhanced Strength Pareto Evolutionary Algorithm 2 is employed. A fuzzy tool is used in addition for selecting the best solution. The main challenge in real-world multi-objective problem is that it requires high computational simulations. The conventional Multi-Objective Evolutionary Algorithms (MOEAs) may take thousands of function evaluation which is not practical for real-world problems. Hence surrogates are introduced in those MOEAs

to approximate the fitness evaluations. Classification of MOEA with respect to their integration with surrogate models is presented in Díaz et al. (Javadian and Haghifam 2008). Yu et al. (2018) proposed a method where PSO and Social learning (SL)-PSO are combined to reach global optimum. To increase the optimization efficiency, a surrogate assisted PSO is proposed in Cai et al. (2019). The algorithm uses global and personal surrogates to update the velocities of the particles. The optimization algorithm plays vital role in all research areas. Selvi et al. (2020) used genetic algorithm for image de-noising process. Senthilselvi et al. (2020) used gravitational search algorithm for image restoration process.

The optimal over current relay coordination in power distribution system is discussed Chen et al. (2013). The type of curve, current setting Tapings and Time dial settings are decided by an experienced engineer to arrive at correct settings of relay. More time and cost are needed for this procedure. The constraints on the power system are reviewed by the engineer for deciding CSM and TSM. The constraints may be maximum fault current, the load current, fault clearing Time and CT ratios. Since a simple software packages do not work with the complex problems, AI techniques such as GA, PSO are implemented to solve the relay coordination problem. Setting the relay operating time and the relay coordination are complicated as the power system encounters various types of loads on different buses, which includes static capacitors, large induction motor, power fuses and transforms. OCR is connected to the CT, the pickup current tuned using CSM and trip time tuned using time dial setting multiplier. The overall operating time of all key relays is minimized at maximum Fault current. The operating curves are simulated in accordance with the IEC for the required operating time and current range of each DOCR. DOCR coordination of a real-time distribution system with embedded Renewable Generation is proposed in Kamal et al. (2014). The relays fail to operate when there is large outage time and huge commercial losses in power plants. The operating time of OC relays is set based on an approximate equivalent circuit of power grid to estimate the short circuit currents. A real-time distribution system with Embedded Renewable Generation is taken. Plug setting and Time dial setting and type of curve are optimized to give better coordination.

Problem formulation

The DOCR relay coordination is a multi-objective problem with objectives of relay operating time and coordination time. The optimal values of TMS and I_p of each relay are computed to arrive at better protection coordination.

Relay characteristics

The time of operation of an extremely inverse type relay is shown in Eq. (1).

$$t_i = \left[\frac{K}{\left(\frac{I_{sci}}{I_{pi}} \right)^\alpha - 1} + L \right] TMS_i \quad (1)$$

In which, current of the short circuit is represented by I_{sci} and i th relay's current pickup setting is denoted by I_{pi} . The range of TMS values is between 0 and 1. The constants K , L and α are 28.2, 0.1217 and 2 as described by standard ANSI/IEEE C37.112–1996 for extremely inverse type relay.

Coordination constraints

Protection coordination is that design of relays such that the primary relay operates first and in case of its failure the back relay comes into action. The coordination time interval (CTI) is time gap between the primary and backup relay. This CTI depends on type of relay, Circuit Breaker speed, microprocessor or electromechanical parts of relay, overshoot interval of the relay and the additional parameters of the system. The CTI is given by:

$$\Delta t_{pb} = t_b - t_p - CTI \quad (2)$$

For each fault location, there may be more than one back relays and with several constraints to be satisfied. So the below condition is defined

$$t_{cij}^{b_k} - t_{cj}^p > CTI \quad (3)$$

where, k is the backup relay identifier, b_k refers to backup relay k , c is the system configuration identifier, $c = 1$ for grid connected configuration and $c = 2$ for double configuration, i is the fault location identifier, j is the relay identifier and the superscript p refers to primary relays. The value of CTI lies between 0.1 and 0.5 s. The values of Time Multiplier Setting (TMS) and pick up current (I_p) of a relay should be also set. The minimum value of pick up current is greater than rated load current. So the following constraints are defined.

$$I_{pi \min} \leq I_{pi} \leq I_{pi \max} \quad (4)$$

$$TMS_{i \min} \leq TMS_i \leq TMS_{i \max} \quad (5)$$

The values of TDS range from 0 to 1. The optimization is carried out for two system configuration, the grid connected configuration and double configuration.

Objective function

With the operating time of the DOCR and the CTI, an objective function is formulated using the overall key relay's operation interval and the overall coordination intervals of the relay to find the best values of TMS's and I_p 's. The objective function is depicted as in Eq. (6).

$$T = \min \sum_{c=1}^C (\alpha_1 \sum_{i=1}^N t_{ci}^q + \alpha_2 \sum_{j=1}^P (\Delta t_{cpb} - \beta_2 (\Delta t_{cpb} - |\Delta t_{cpb}|)))^r \quad (6)$$

where, c is the identifier of mode of operation of micro grid, the i th relay's nearby fault operation interval is denoted as t_{ci} , the synchronization interval between two adjacent relays is represented as Δt_{cpb} , the total count of the available relays is denoted by N , the count of available key and secondary pair of relays are represented as P , the value of i lies between 1 and N , in which i is the relay, α_1 and α_2 are the control weights and non-synchronization parameter is represented as β_2 . The powers raised in the equation are denoted using parameters such as q and r . To attain maximum relay operation interval during both the operation modes of micro grid, the parameters α_1 , α_2 , β_2 , and power values, q and r are also optimized.

Surrogate assisted optimization algorithm

In Engineering, many optimization problems are considered as global optimization problems. Conventional optimization techniques such as Newton method, steepest descent are used to solve many engineering optimization problems. But they have the limitations, that they are not able to find a global solution in case of problems with multiple local minima. Hence the conventional optimization algorithms are replaced by global optimization algorithms. These algorithms are nowadays used in many engineering problems. These algorithms are not knowledge based and provide better accurate results. Even though these optimization algorithms are useful to solve complex optimization algorithms, they are incapable when the objective function is computationally complex and expensive, since it requires high number of iterations and function evaluations. The time taken to complete first iteration is somewhat high in these types of nature inspired evolutionary algorithms and it is not applied to real-world optimization problems with large number of function evaluations. Hence nowadays, surrogate assisted optimization algorithms are used to solve the computationally expensive real-world optimization problems sub optimization technique. The surrogate models are used to improve the search and converge quickly to the global optimum. The surrogate models are built from the random data obtained

in problem space. These models represent the approximation of objective function and constraints. The conventional evolutionary optimization algorithms are used to search the global optimum solution. The computational cost is very low as compared to numerical analysis. The surrogate models are constructed using experimental designs. The surrogate-assisted optimization also suffers from limitations such as choosing of samples, surrogate model building and accuracy of constructed model (Amali and Baskar 2015). Hence surrogate assisted approach is efficient when used with other global optimization techniques.

Surrogate assisted particle swarm optimization algorithm (SAPSO)

Optimization algorithms with great estimating performance can solve the complicated real-time problems. Several algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO) algorithm and Differential evolution (DE) algorithms are implemented to handle complex real-world issues. However these algorithms require great number of fitness evaluations to reach the global optimum solution which is imposes a great challenge for the real-world scenario. The introduction of surrogate models to these algorithms reduces the number of fitness evaluations and is being implemented practically. The polynomial regression, Radial Basis Function Network (RBFN), Artificial Neural Network (ANN), Kriging or Gaussian process (GP) and Support Vector Machines (SVM) are incorporated as the surrogate models (Yu et al. 2018). The SAPSO obtains the Pareto optimal solution in case of multi-objective optimization technique using less number of fitness evaluations. The SAPSO differ from basic PSO in updating mechanism, i.e., it is based on surrogates. The number of fitness evaluations is reduced by approximating the real-valued function, using surrogate models. The surrogate learns a rough contour of the fitness landscape in wide search space. RBFN Radial Basis Function Network is used as approximation models, which is used for function approximation. So in SAPSO the convergence is faster compared.

Surrogate assisted PSO updating mechanism

Particle swarm optimization Taking the number of particles as NP, the real fitness function is computed with an initial set of solutions generated by a Latin Hyper Cube sampling. The initial solutions and the calculated fitness functions are stored in a database. The local and the global optimal values are upgraded. The position and velocity of the particles at t generation are updated using the Eq. (7) and (8)

$$v_i(t+1) = X(v_i(t) + c_1 r_1 (p_i(t) - x_i(t)) + c_2 r_2 (p_g(t) - x_i(t))) \quad (7)$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (8)$$

Where, X indicates the inertia weight, c_1 , c_2 are the learning rate, r_1 , r_2 represent the random numbers which are belong to (0, 1), $p_i(t)$, $p_g(t)$ are the local and the global best at t generation respectively. The main objective of SAPSO is to enhance the cognitive characteristics and social characteristics of the swarm particles. Also, it smoothes the local optima and hence the global surrogate is the optimal value of the whole search space. The SAPSO aims to improve the efficiency of the generic PSO algorithm by updating the velocity by surrogate assistance.

Formation of surrogates

The fitness function is approximated using Gauss process to generate the surrogates. The computationally expensive fitness functions are replaced by the surrogate models.

Then initial position and velocity are updated as

$$\begin{aligned} x'_i(1) &= x_i(t), v'_i(1) = v_i(t), p'_i(1) \\ &= p_i(t), p'_g(1) = p_g(t), v'_i(k+1) \\ &= wv'_i(k) + c_1 r_1 [p'_i(k) - x'_i(k)] \\ &\quad + c_2 r_2 [p'_g(k) - x'_i(k)] \end{aligned} \quad (9)$$

$$x'_i(k+1) = x'_i(k) + v'_i(k+1) \quad (10)$$

The velocity and the position of the i th particle in the virtual space at the k th iteration are $v'_i(k)$ and $x'_i(k)$. The local and the global best of the swarm at the kth iteration in the virtual space are $p'_i(k)$ and $p'_g(k)$. A' and A are the external archives in the virtual space and the real physical space respectively.

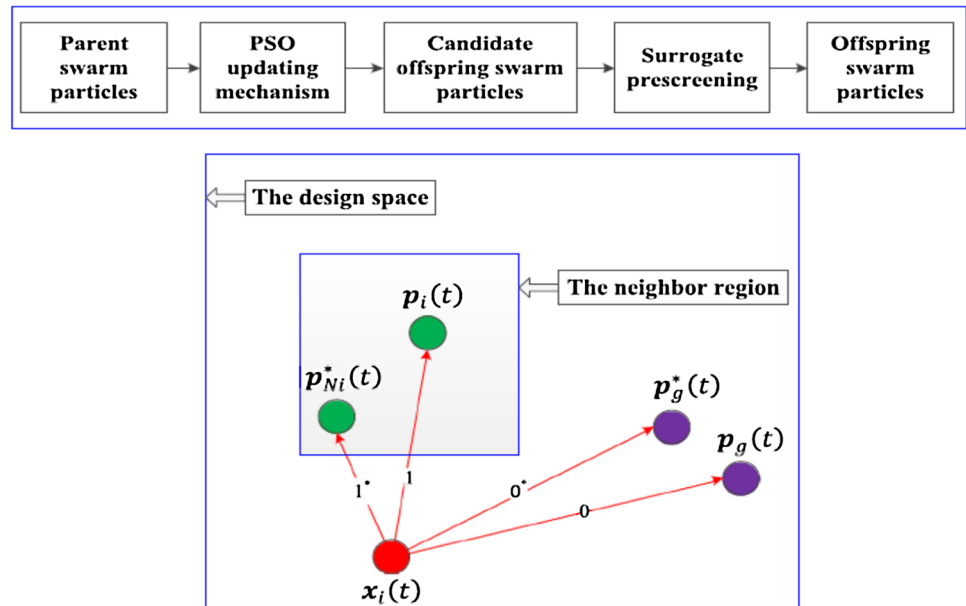
Particles deviated from the pareto front

The particles with the non-dominant solutions in database are taken as reference front. The Euclidean distances are calculated between the fitness value of the particles and the reference front. The particles whose Euclidean distance is larger than average distance are taken to be deviated from Pareto front.

Let x'_i is the particle deviated from the Pareto front, and then:

$$x''_i(k) = x'_i(k) + r[x'_i(k) - x^*_i(k)] \quad (11)$$

Fig. 1 SAPSO updating mechanism



where, $x_i''(k)$ indicates the new position of the i th particle in the virtual space at the k th iteration; x_i^* denotes the closest non-dominant solution to x' ; $r \in \{0,1\}$.

Particles scattered from pareto front

The two nearest non-dominant solutions are considered and their Euclidean distance is calculated. When their Euclidean distance is greater than the average distance of all adjacent non-dominant solutions, then the particles are to be scattered.

Let x_h^* and x_{h+1}^* are particles scattered of the Pareto front. Then the new position of the particles are calculated by

$$x'''(k) = \begin{cases} x_h^*(k) + r(x_{h+1}^*(k) - x_h^*(k)) \\ x_{h+1}^*(k) + r(x_h^*(k) - x_{h+1}^*(k)) \end{cases} \quad (12)$$

where $x'''(k)$ indicates the position of new particle scattered of the Pareto front in virtual space at k th iteration.

The final set of solution is obtained from

$$X_{pss} = \{x'(k), x''(k), x'''(k)\}; k = 1 : \eta \quad (13)$$

where X_{pss} is final set of solutions; x' , x'' , x''' indicate the solutions calculated from the above operations.

From the solutions obtained, the solutions are classified as Pareto-optimal and non-Pareto-optimal solutions with high probability. The process is terminated when all the candidate solutions are classified. The approximate Pareto solution set in the external database is identified and the global best solution is generated to enhance the social experience of the population. The Fig. 1 shows the surrogate updating mechanism in PSO.

Table 1 Parameters of SAPSO

Parameters	Specification
Population size, N_p	50
Inertia weight ω	0.7
Learning rate $C1$, $C2$	1, 2
Learning factor τ	0.02
Prediction accuracy ϵ	$\{0, 1\}$
Classification cost	5
No. of greedy searches	1000

Pseudocode of SAPSO

Step 1: A population of particles is initialised using sampling methods. The particles are assessed by a fitness function and the values are updated in exterior files and databases.

Step 2: The best value of the particles and the global best value of the population of the particles are recorded.

Step 3: An approximate model is constructed using the information in the data base. The fitness function is modified using the approximate model for each objective.

Step 4: Sampling is performed to obtain the solutions and are separated as Pareto and non-Pareto solutions.

Step 5: The solutions are updated in exterior files and the procedure is repeated until the number of fitness evaluation reaches maximum value (Olivares et al. 2014).

The key feature in SAPSO is the sampling and the classification of the solutions (Olivares et al. 2014). The parameters for SAPSO are shown in Table 1.

Results and discussion

The proposed SAPSO algorithm is evaluated with IEEE 6 bus and IEEE 30 bus system for both grids connected and double configurations. SAPSO based protection scheme is explained as flow chart in Fig. 2. All simulations are carried out in MATLAB environment and the analysis is performed. In grid connected configuration, the system is supplied from grid and DG also. In Double configuration, the constraints includes both grid connected and islanded mode, i.e., only DG presents. The optimization is carried out first for grid connected mode. Then the two modes are considered simultaneously and the optimum relay setting is obtained. In this case, the relay can satisfactorily work in both modes of operation of microgrid.

Case 1: IEEE 6 bus system

The SAPSO is tested with IEEE 6 bus system. It consists of 7 lines, 3 generators and 14 relays. Each generator is 127.5 MW rating. Each line is introduced with midway points for indicating where faults are created. Each fault position is associated with one primary relay and two or three back up relays. The aim of SAPSO is to find the TMS and pick up current value by proper settings of relays at least computational cost. Three phase faults are created in the lines. The algorithm computes the optimal value of TMS and pick up current by proper settings of primary relays. The CT ratios of the relays are presented in Table 2.

Grid connected configuration

The values of TMS and pick up current (I_p) are optimized and are shown in the Table 3. The relay operating times were obtained with best values of TMS, I_p using SAPSO algorithm. The value of total TMS of all the relays is computed as 1410 milli seconds and pickup current as 1760 milli amperes. The TMS values of all the relays R1 to R14 are within 0 and 1. Also the pickup current values of all relays are optimized according to load currents.

Double configuration

The values of TMS and I_p for grid connected portion of double configuration considering both grid connected mode and islanded mode simultaneously is shown in Table 4.

The TMS values of all the relays R1–R14 are within 0 and 1. Also the pickup current I_p values of all relays are optimized according to load currents. Table 5 shows the optimal settings for all relays in double configuration. The best values of TMS, I_p are computed using SAPSO Algorithm.

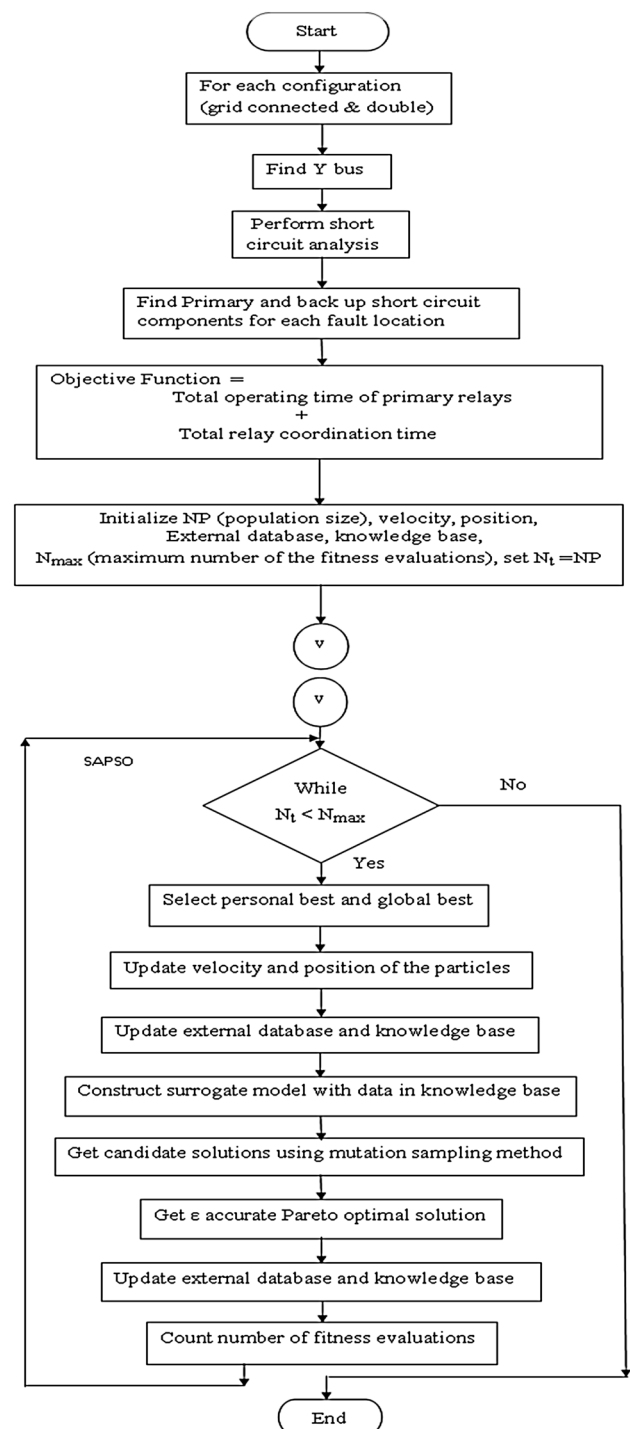


Fig. 2 Flow chart explaining SAPSO based protection scheme for micro grids

The value of total TMS is computed as 1510 ms and pickup current as 1920 mA for dual configuration.

As shown in Table 5, the total operating time of all primary relays is calculated as 9884 ms for single configuration and 12,192 ms for grid connected portion of dual

Table 2 CT Ratio for 6-bus test system

Relay no	CT ratio	Relay no	CT ratio
1	1200/5	8	800/5
2	800/5	9	800/5
3	800/5	10	600/5
4	800/5	11	800/5
5	800/5	12	800/5
6	1200/5	13	1200/5
7	800/5	14	800/5

Table 3 Time multiplier setting and pick up current for grid connected configuration using SAPSO

TMS no	TMS (ms)	$I_{p\ no}$	I_p (mA)
TMS ₁	80	I_{p1}	130
TMS ₂	90	I_{p2}	160
TMS ₃	40	I_{p3}	150
TMS ₄	120	I_{p4}	120
TMS ₅	200	I_{p5}	300
TMS ₆	30	I_{p6}	100
TMS ₇	70	I_{p7}	160
TMS ₈	90	I_{p8}	0
TMS ₉	100	I_{p9}	120
TMS ₁₀	140	I_{p10}	140
TMS ₁₁	220	I_{p11}	120
TMS ₁₂	110	I_{p12}	80
TMS ₁₃	50	I_{p13}	20
TMS ₁₄	70	I_{p14}	160
Total TMS (ms)	1410	Total I_p (mA)	1760

Table 4 Time multiplier setting and pickup current for grid connected portion of double configuration using SAPSO

TMS no	TMS (ms)	I_p no	I_p (mA)
TMS ₁	70	I_{p1}	100
TMS ₂	50	I_{p2}	150
TMS ₃	70	I_{p3}	130
TMS ₄	60	I_{p4}	170
TMS ₅	50	I_{p5}	330
TMS ₆	100	I_{p6}	130
TMS ₇	70	I_{p7}	140
TMS ₈	90	I_{p8}	40
TMS ₉	200	I_{p9}	130
TMS ₁₀	140	I_{p10}	140
TMS ₁₁	160	I_{p11}	150
TMS ₁₂	120	I_{p12}	120
TMS ₁₃	190	I_{p13}	30
TMS ₁₄	140	I_{p14}	160
Total TMS (ms)	1510	Total I_p (mA)	1920

Table 5 Total operating time of primary relays for IEEE 6 bus system

Relay	Single configuration Time of operation (ms)	Double configuration
Time ₁	467	575
Time ₂	678	753
Time ₃	756	823
Time ₄	900	1000
Time ₅	245	360
Time ₆	1137	1352
Time ₇	145	120
Time ₈	351	453
Time ₉	1345	1563
Time ₁₀	524	754
Time ₁₁	536	742
Time ₁₂	1463	1893
Time ₁₃	354	564
Time ₁₄	983	1240
Total time (ms)	9884	12,192

configuration. Thus the operating time obtained considering both constraints simultaneously matches results of those obtained from single configuration. Hence the SAPSO gives the optimal settings of TMS and pick up current for the IEEE 6 bus system ensuring relay coordination and also satisfying the CTI. By analyzing the two modes of operation of micro grid simultaneously, the feasible solution for settings of relay is found. It is observed from the results shown in Table 6, that each primary relay have backup relays for efficient operation. The relays are co-ordinate correctly. For example the coordination time between R11 and R7 is 99.23 ms under grid connected configuration and 106.62 ms under double configuration. The coordination time is within 200 ms. And also coordination time between R8 and R1 is only 20.23 ms. Since the relay R8 is connected to generator bus. This reveals that the relay is most sensitive and so, the coordination time is very less.

Case 2: IEEE 30 bus system

The protection scheme is tested on IEEE 30 bus system. The system has 2 subsystems a 132 kV sub transmission system and a 33 kV distribution system with 78 relays.

The distribution system is analyzed. It has 37 lines, 6 Generators, 4 transformers and 78 DOCR. The aim of SAPSO is to find the TMS and pick up current value by proper allocation of relays. Three phase faults are created in the line. The algorithm computes the optimal value of TMS and pick up current by proper arrangements of primary relays. The TMS and pick up current optimized are listed in the Table 7.

Table 6 Coordination time for primary and back up relay pairs

$\Delta time_{pb}$	Coordination time for primary and back up relay pairs (ms)	
	Single configuration	Grid connected mode of double configuration
$\Delta time_{8,1}$	20.33	25.31
$\Delta time_{1,4}$	132.2	138
$\Delta time_{1,13}$	170.26	176.54
$\Delta time_{2,7}$	20.1	18.46
$\Delta time_{6,13}$	42.3	49.56
$\Delta time_{4,14}$	28.2	31.52
$\Delta time_{4,9}$	46	50.64
$\Delta time_{10,14}$	10.2	12.34
$\Delta time_{10,11}$	20.3	19.25
$\Delta time_{13,9}$	26.4	29.8
$\Delta time_{5,2}$	76.4	85.2
$\Delta time_{5,1}$	124.6	130
$\Delta time_{11,7}$	99.23	106.62
$\Delta time_{11,1}$	66.34	73.39
$\Delta time_{12,13}$	89.4	98.64
$\Delta time_{12,5}$	46.5	44.65

The TMS values of all the relays lies within 0 and 1. The best value of total TMS is 37229 ms and pickup current is 8759 mA. The relay operating times were shown in Table 8 obtained with best values of TMS, I_p using SAPSO algorithm. The total operating time of all primary relays is calculated as 49,077 ms. So, with the obtained settings of TMS and pick up current, the relays in the IEEE 30 bus system operate with proper coordination satisfying CTI.

Comparison of multi objective particle swarm optimization (MOPSO) With SAPSO

The IEEE 9 bus system is tested with SAPSO and the results are compared with MOPSO. From the Table 9, it is inferred that the total operating time for MOPSO for single configuration is 14023.2 ms while for SAPSO is 10522 ms.

Similarly for double configuration SAPSO provides low operating time of 11,504 ms compared to MOPSO. The operating time of R15 under single configuration is 145 ms when SAPSO is applied and it is 165 ms when MOPSO is applied. Hence SAPSO offers encouraging results in terms of operating time. All relays are correctly coordinated. The CTI is within the 200 ms and this reveals the optimal coordination between the relays.

The Inverted Generation Distance (IGD) (Lv et al. 2019) is employed as the parameter for computing the performance of the algorithm with 500 points sampled. The IGD is defined as

$$IGDof(P_{true} + P_{rep}) = \sum_{x \in P_{true}} \frac{dis(x, P_{rep})}{|P_{true}|} \quad (14)$$

where P_{true} are the true Pareto Front, P_{rep} represents the non-dominant solution in the external archive, $|P_{true}|$ represents the number of real forward reference points. The quality of the solution is indicated by the IGD in terms of location of the solution and arrival at the solution.

The least IGD offers better quality of solution. The standard MOPSO and SAPSO algorithms are compared in terms of arithmetic mean and diverse values of IGD for the multi objective coordination problem with 500 function evaluations and the TMS and I_p are obtained for both single grid connected configuration and the double configuration. The Table 10 shows that the mean and the standard deviation of the IGD offered by the SAPSO are minimum when compared to MOPSO. The minimum value of SD indicates that the SAPSO have good stability than MOPSO.

To have clear vision of the convergence and distribution of the SAPSO and MOPSO the comparison graph for 500 function evaluations is presented in Fig. 3. It is proved that the SAPSO arrives to the true Pareto front and yields the distinct solutions when compared to MOPSO. The introduction of surrogates approximates the fitness function and offers better Pareto fronts, convergence and distributions. The optimum operating time of the relay can be obtained at 342 iterations in SAPSO and 356 iterations in MOPSO.

The performance of the SAPSO is evaluated using the mean of IGD. The IGD of MOPSO and SAPSO for various number of function evaluations 500, 1000, 1500 and 2000 is presented in Fig. 4. SAPSO have better performance than the MOPSO in arriving at the solution with same number of fitness evaluations.

Conclusion

A SAPSO approach for the optimal coordination of DOCRs for both grids connected and double configuration was presented. Coordination problem is solved using multi-objective approach. The CTI was taken as one of the objective along the operating time of all relays and investigated considering grid and autonomous mode constraints simultaneously. The protection scheme was tested using standard IEEE 6 and 30 bus system. For the IEEE 6 bus system the optimal settings of TMS and I_p are computed so that the relay employed operates satisfactorily, when micro grid is in grid connection mode or it has to perform in islanded mode. The total operating time of all primary relays is calculated as 9884 ms for single configuration and 12,192 ms for grid connected portion of double configuration. Thus the operating time obtained considering both constraints simultaneously

Table 7 Time multiplier setting and pick up current using SAPSO

TMS no	TMS (ms)	I_P no	I_P (mA)	TMS no	TMS (ms)	I_P no	I_P (mA)
TMS ₁	111	I_{P1}	108	TMS ₁₅	15	I_{P15}	139
TMS ₂	780	I_{P2}	153	TMS ₁₆	43	I_{P16}	166
TMS ₃	390	I_{P3}	141	TMS ₁₇	169	I_{P17}	109
TMS ₄	242	I_{P4}	117	TMS ₁₈	649	I_{P18}	45
TMS ₅	404	I_{P5}	22	TMS ₁₉	732	I_{P19}	33
TMS ₆	96	I_{P6}	142	TMS ₂₀	648	I_{P20}	89
TMS ₇	132	I_{P7}	175	TMS ₂₁	451	I_{P21}	28
TMS ₈	942	I_{P8}	118	TMS ₂₂	547	I_{P22}	39
TMS ₉	956	I_{P9}	117	TMS ₂₃	296	I_{P23}	41
TMS ₁₀	575	I_{P10}	42	TMS ₂₄	745	I_{P24}	176
TMS ₁₁	60	I_{P11}	132	TMS ₂₅	189	I_{P25}	107
TMS ₁₂	235	I_{P12}	117	TMS ₂₆	687	I_{P26}	43
TMS ₁₃	353	I_{P13}	66	TMS ₂₇	184	I_{P27}	84
TMS ₁₄	821	I_{P14}	109	TMS ₂₈	368	I_{P28}	125
TMS ₂₉	626	I_{P29}	82	TMS ₅₄	471	I_{P54}	193
TMS ₃₀	780	I_{P30}	182	TMS ₅₅	230	I_{P55}	195
TMS ₃₁	81	I_{P31}	196	TMS ₅₆	844	I_{P56}	75
TMS ₃₂	929	I_{P32}	163	TMS ₅₇	195	I_{P57}	148
TMS ₃₃	776	I_{P33}	135	TMS ₅₈	226	I_{P58}	124
TMS ₃₄	487	I_{P34}	170	TMS ₅₉	171	I_{P59}	106
TMS ₃₅	436	I_{P35}	190	TMS ₆₀	228	I_{P60}	92
TMS ₃₆	447	I_{P36}	136	TMS ₆₁	436	I_{P61}	34
TMS ₃₇	306	I_{P37}	136	TMS ₆₂	311	I_{P62}	67
TMS ₃₈	509	I_{P38}	170	TMS ₆₃	923	I_{P63}	51
TMS ₃₉	511	I_{P39}	46	TMS ₆₄	430	I_{P64}	138
TMS ₄₀	818	I_{P40}	121	TMS ₆₅	185	I_{P65}	89
TMS ₄₁	795	I_{P41}	71	TMS ₆₆	905	I_{P66}	197
TMS ₄₂	644	I_{P42}	172	TMS ₆₇	980	I_{P67}	61
TMS ₄₃	379	I_{P43}	45	TMS ₆₈	439	I_{P68}	102
TMS ₄₄	812	I_{P44}	48	TMS ₆₉	111	I_{P69}	28
TMS ₄₅	533	I_{P45}	61	TMS ₇₀	258	I_{P70}	164
TMS ₄₆	351	I_{P46}	104	TMS ₇₁	409	I_{P71}	121
TMS ₄₇	939	I_{P47}	106	TMS ₇₂	595	I_{P72}	99
TMS ₄₈	876	I_{P48}	151	TMS ₇₃	262	I_{P73}	62
TMS ₄₉	550	I_{P49}	168	TMS ₇₄	603	I_{P74}	32
TMS ₅₀	622	I_{P50}	33	TMS ₇₅	711	I_{P75}	126
TMS ₅₁	587	I_{P51}	127	TMS ₇₆	222	I_{P76}	185
TMS ₅₂	208	I_{P52}	185	TMS ₇₇	121	I_{P77}	128
TMS ₅₃	301	I_{P53}	161	TMS ₇₈	840	I_{P78}	200
				TMS (ms)	37,229	I_P (mA)	8759

matches results of those obtained from single configuration. Each primary relay has backup relays for efficient operation. The coordination time is also within 200 ms. This reveals the optimum coordination between the primary and backup relays.

Considering IEEE 30 bus system the total operating time of all primary relays is calculated as 49,077 ms. So with the obtained settings of TMS and pick up current, the relays in

the IEEE 30 bus system operate with proper coordination satisfying CTI. Also the IEEE 9 bus system is tested with SAPSO and compared with MOPSO. The total operating time for MOPSO for single configuration is 14023.2 ms while for SAPSO is 10522 ms. Similarly for double configuration SAPSO provides low operating time of 11,504 ms compared to MOPSO. The operating time of R15 under single configuration is 145 ms when SAPSO is applied and it

Table 8 Operating time of primary relay for IEEE 30 bus system

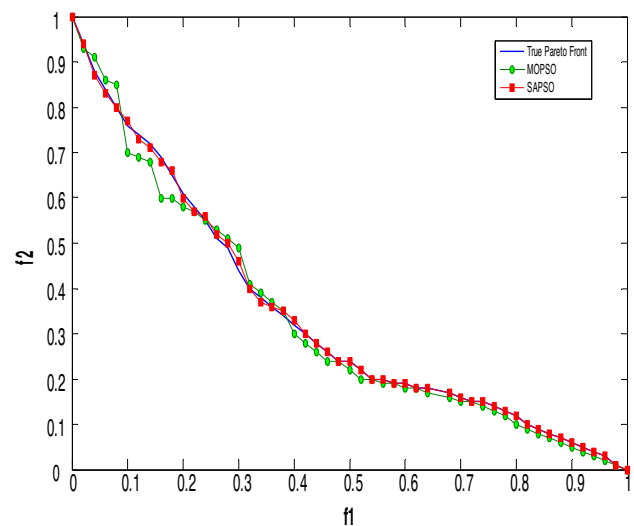
Primary relay	Operating time single configura- tion (ms)	Primary relay	Operating time single configura- tion (ms)	Primary relay	Operating time single configura- tion (ms)	Primary relay	Operating time single configura- tion (ms)
1	688	21	663	41	575	61	422
2	865	22	912	42	453	62	912
3	391	23	999	43	333	63	475
4	798	24	850	44	434	64	334
5	875	25	481	45	596	65	201
6	883	26	356	46	505	66	887
7	974	27	662	47	807	67	816
8	842	28	960	48	930	68	456
9	618	29	322	49	297	69	694
10	917	30	206	50	280	70	483
11	424	31	529	51	243	71	554
12	945	32	605	52	943	72	968
13	619	33	929	53	254	73	220
14	519	34	742	54	987	74	315
15	675	35	219	55	932	75	633
16	777	36	457	56	648	76	568
17	943	37	899	57	748	77	900
18	505	38	413	58	595	78	340
19	253	39	420	59	564	Total time (ms) 49,077	
20	940	40	854	60	846		

Table 9 Comparison of MOPSO with SAPSO

Relay	Operating time Single configuration(ms)		Operating time Double configuration(ms)	
	MOPSO	SAPSO	MOPSO	SAPSO
Time ₁	596	493	600	595
Time ₂	997	813	1000	675
Time ₃	600.2	703	620	113
Time ₄	885	263	1000	213
Time ₅	365.4	124	380	294
Time ₆	1267	119	1460	930
Time ₇	113	813	120	345
Time ₈	2185	2037	2200	2133
Time ₉	685	214	760	895
Time ₁₀	1127.5	703	1140	1011
Time ₁₁	589.6	625	570	412
Time ₁₂	1368	1313	1480	1390
Time ₁₃	367.5	413	380	199
Time ₁₄	1245	1132	1300	833
Time ₁₅	165	145	140	113
Time ₁₆	1467	612	1560	1353
Total time(ms)	14,023.2	10,522	14,710	11,504

Table 10 Comparison of SAPSO with MOPSO

Parameter	MOPSO	SAPSO
Mean	0.279	0.0672
S.D	0.007	0.003

**Fig. 3** Comparison of convergence obtained by SAPSO and MOPSO

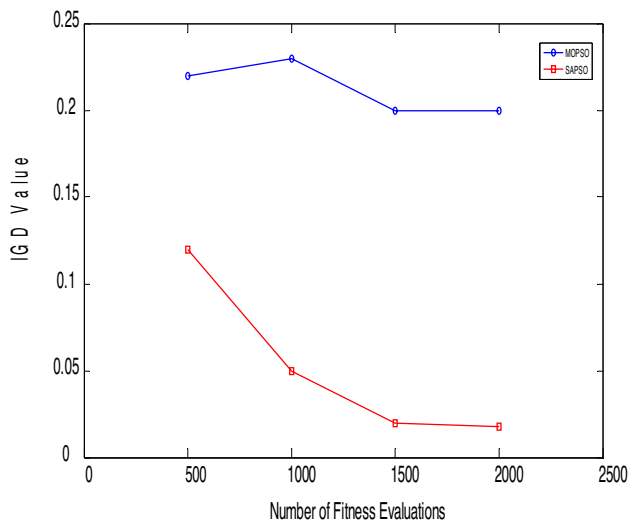


Fig. 4 Comparison of convergence obtained by SAPSO and MOPSO

is 165 ms when MOPSO is applied. Hence SAPSO offers encouraging results in terms of operating time. The CTI is within the 200 ms and this reveals the optimal coordination between the relays.

The performance of the SAPSO is evaluated by comparison of the mean of IGD of MOPSO and SAPSO. The mean and the standard deviation of the IGD offered by the SAPSO are minimum when compared to MOPSO. The minimum value of SD indicates that the SAPSO have good stability than MOPSO. From the convergence characteristic analysis it is proved that SAPSO offers better Pareto fronts, convergence and distributions. The mean of IGD obtained for various function evaluations namely 500, 1000, 1500 and 2000 are compared using graphical analysis. SAPSO offers effective results than the MOPSO in arriving at the solution. The introduction of surrogates approximates the fitness function and offers better Pareto Fronts, convergence and distribution.

Future scope of the work

Surrogate models in transformative calculations, can have both pros and cons effects. Erroneous surrogates may lead evolutionary algorithms to a deviation from accurate results. So neighbourhood surrogate and well as a global surrogate are needed. The neighbourhood surrogate model may serve to smooth out local optimum, and hence accelerating the search. The smoothing impact of the global surrogate can make the search to be a quicker than on the original fitness function. By properly combining the global surrogate with local surrogate models, the global optimum

can be attained rapidly and accurately. So a two-layer surrogate model is predicted to apply in future for fast and accurate results.

Declarations

Conflict of interest The authors declare they have no conflicts of interest.

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